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Set-10

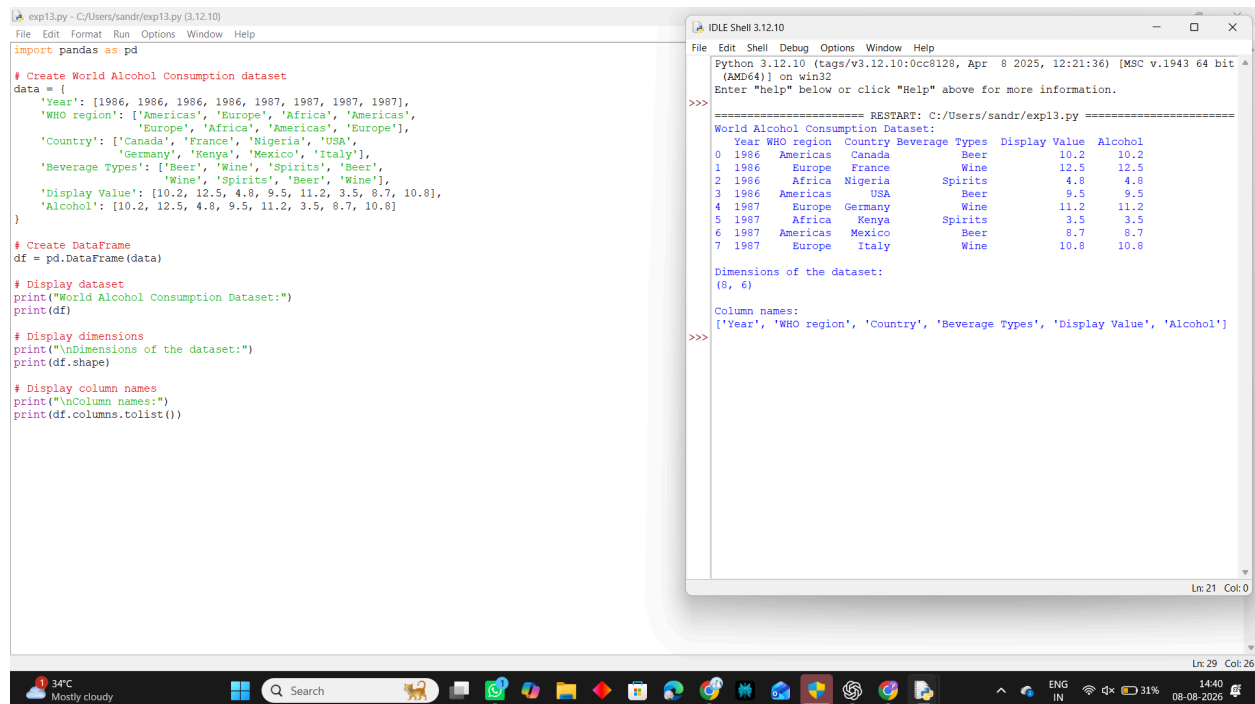
19. Write a Pandas program to display the dimensions or shape of the World alcohol consumption dataset. Also extract the column names from the dataset.

Aim

To write a Pandas program to display the **dimensions (shape)** of the World Alcohol Consumption dataset and extract the **column names** from the dataset.

Procedure

1. Import the Pandas library.
2. Read the World Alcohol Consumption dataset using `pd.read_csv()`.
3. Use the `shape` attribute to find the number of rows and columns.
4. Use the `columns` attribute to extract the column names.
5. Display the results.



```
exp13.py - C:/Users/sandr/exp13.py (3.12.10)
File Edit Format Run Options Window Help

import pandas as pd

# Create World Alcohol Consumption dataset
data = {
    'Year': [1986, 1986, 1986, 1986, 1987, 1987, 1987, 1987],
    'WHO region': ['Americas', 'Europe', 'Africa', 'Americas',
                  'Europe', 'Africa', 'Americas', 'Europe'],
    'Country': ['Canada', 'France', 'Nigeria', 'USA',
               'Germany', 'Kenya', 'Mexico', 'Italy'],
    'Beverage Types': ['Beer', 'Wine', 'Spirits', 'Beer',
                      'Wine', 'Spirits', 'Beer', 'Wine'],
    'Display Value': [10.2, 12.5, 4.8, 9.5, 11.2, 3.5, 8.7, 10.8],
    'Alcohol': [10.2, 12.5, 4.8, 9.5, 11.2, 3.5, 8.7, 10.8]
}

# Create DataFrame
df = pd.DataFrame(data)

# Display dataset
print("World Alcohol Consumption Dataset:")
print(df)

# Display dimensions
print("\nDimensions of the dataset:")
print(df.shape)

# Display column names
print("\nColumn names:")
print(df.columns.tolist())
```

```
IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/sandr/exp13.py =====
World Alcohol Consumption Dataset:
  Year WHO region Country Beverage Types Display Value Alcohol
0 1986   Americas   Canada      Beer      10.2      10.2
1 1986   Europe    France      Wine      12.5      12.5
2 1986   Africa  Nigeria    Spirits       4.8       4.8
3 1986   Americas    USA      Beer       9.5       9.5
4 1987   Europe   Germany      Wine      11.2      11.2
5 1987   Africa    Kenya    Spirits       3.5       3.5
6 1987   Americas  Mexico      Beer       8.7       8.7
7 1987   Europe    Italy      Wine      10.8      10.8

Dimensions of the dataset:
(8, 6)

Column names:
['Year', 'WHO region', 'Country', 'Beverage Types', 'Display Value', 'Alcohol']
>>>
```

Result

Thus, the World Alcohol Consumption dataset was successfully created, and its **dimensions and column names** were displayed using Pandas.

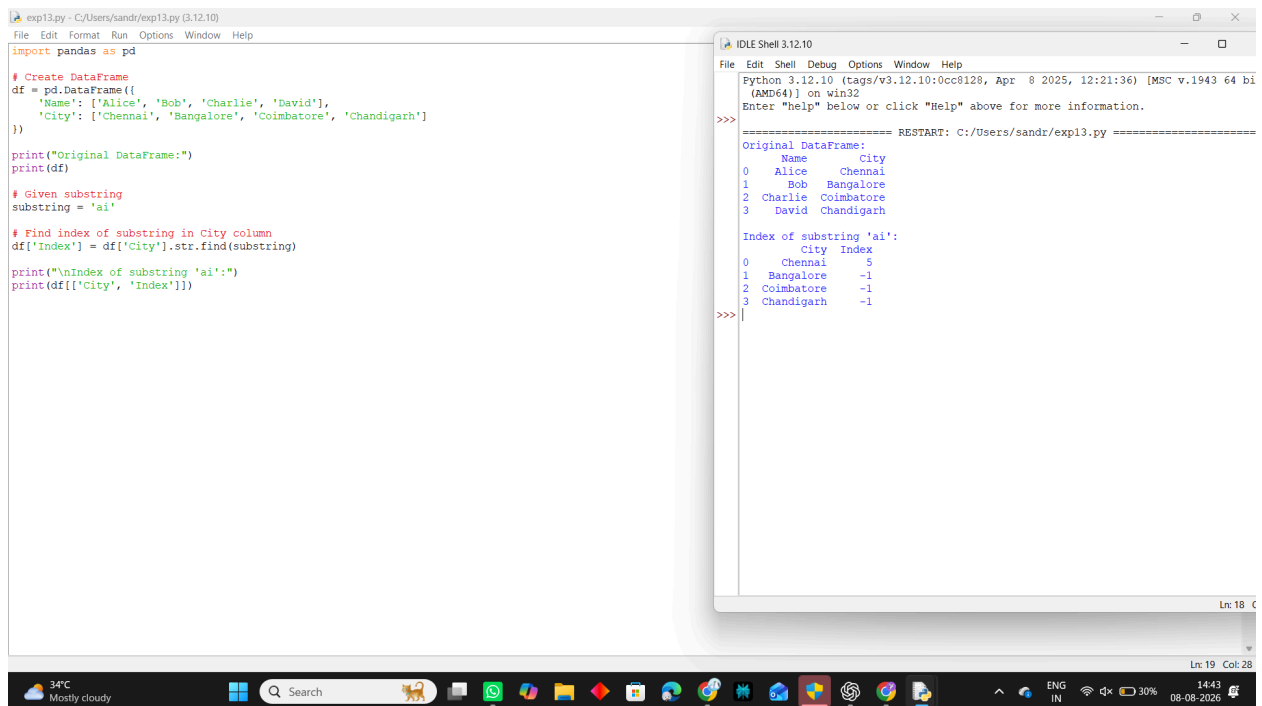
20. Write a Pandas program to find the index of a given substring of a DataFrame column.

Aim

To write a Pandas program to find the **index position of a given substring** in a DataFrame column.

Procedure

1. Import the Pandas library.
2. Create a DataFrame containing a text column.
3. Specify the substring to search for.
4. Use the `str.find()` function to find the position of the substring.
5. Display the result.



The screenshot shows a Python IDE with two windows. The left window displays the source code for a script named `exp13.py`. The code imports the `pandas` library, creates a DataFrame with names and cities, and uses the `str.find()` method to find the index of the substring 'ai' in the 'City' column. The right window shows the output of the script, which prints the original DataFrame and a new DataFrame containing the index of the substring 'ai' for each city.

```
import pandas as pd

# Create DataFrame
df = pd.DataFrame({
    'Name': ['Alice', 'Bob', 'Charlie', 'David'],
    'City': ['Chennai', 'Bangalore', 'Coimbatore', 'Chandigarh']
})

print("Original DataFrame:")
print(df)

# Given substring
substring = 'ai'

# Find index of substring in City column
df['Index'] = df['City'].str.find(substring)

print("\nIndex of substring 'ai':")
print(df[['City', 'Index']])
```

Output:

```
Original DataFrame:
   Name  City
0  Alice  Chennai
1   Bob  Bangalore
2 Charlie  Coimbatore
3  David  Chandigarh

Index of substring 'ai':
   City  Index
0  Chennai     5
1  Bangalore    -1
2  Coimbatore    -1
3  Chandigarh    -1
```

Result

Thus, the Pandas program was successfully executed to find the **index position of the given substring** in a DataFrame column.